

SATNET EDU / Lecture 03

Links, Topology, and Routing

How usable links become end-to-end paths through a satellite network.

LECTURE 03

From physical visibility to a routing decision

Build the graph. Choose an objective. Explain the route.

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A Route Needs Both Ground and Space Links

Coverage supplies a possible first hop. Reaching another terminal also requires a connected path.

► **Ground links enter and leave space.**

- A terminal uses a ground–satellite link. A gateway can connect a satellite network to terrestrial infrastructure.

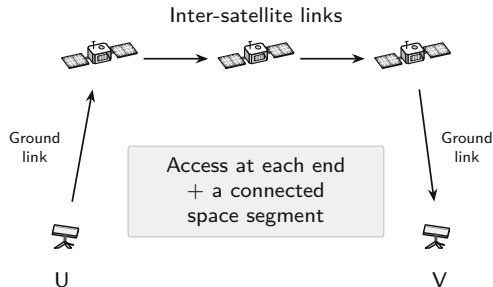
► **Space links join satellites.**

- An inter-satellite link (ISL) carries traffic between satellites. Radio or optical terminals must establish that link.

► **Every hop must be usable.**

- Two covered ground sites can still be disconnected when their satellites have no connecting chain.

Trace a path from terminal U to terminal V



Generic architecture. The SatNet Edu ground stations are endpoints, not terrestrial transit routers.

A Nearby Satellite Can Still Be Hidden by Earth

An ISL candidate needs a clear line of sight as well as an acceptable distance.

► Test the entire line segment.

- The straight segment joining two satellites must stay outside Earth. A short distance alone cannot establish visibility.

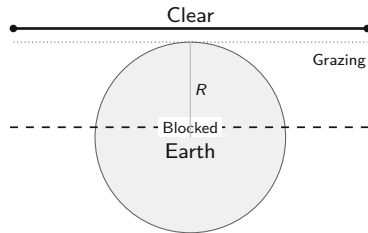
► Altitude sets the grazing limit.

- For equal orbital radii $r = R + h$, the grazing separation is $d_{\max} = 2\sqrt{r^2 - R^2}$.

► Geometry is only one filter.

- Hardware range, pointing, terminal availability, and scheduling can remove otherwise visible links.

Clear, blocked, and grazing links



At 550 km: grazing distance ≈ 5408 km

Spherical Earth, $R = 6371$ km. The simulator blocks exact tangency. Diagram is schematic.

A Ground Link Gets Longer Near the Horizon

The same orbit gives different link distances as a satellite crosses the terminal's sky.

► **Elevation determines slant range.**

- The slant range d is the straight terminal-to-satellite distance. It equals altitude only for an overhead satellite.

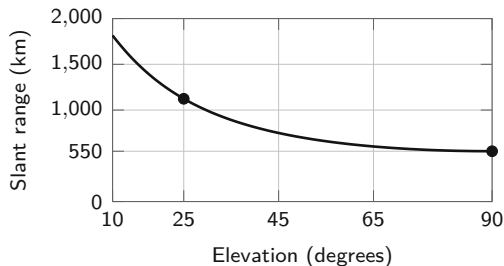
► **Distance sets propagation time.**

- For a space radio link, $t_p = d/c$. A 550 km overhead link takes about 1.83 ms in one direction.

► **A mask trades reach for geometry.**

- Raising the minimum elevation removes longer, low-angle links and reduces the footprint established in Lecture 02.

550 km altitude, one ground link



$$d = \sqrt{(R + h)^2 - R^2 \cos^2 e} - R \sin e$$

$R = 6371$ km, $h = 550$ km, elevation e , $c = 299792.458$ km/s. Propagation only.

A Topology Policy Chooses Which Links to Test

A physically possible pair is not automatically an installed or scheduled connection.

► **Start with candidate pairs.**

- A regular constellation can nominate neighbors along each orbital plane and across nearby planes.

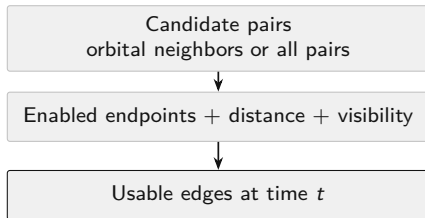
► **Apply geometry and status filters.**

- A candidate becomes a usable edge only when its endpoints are enabled, its range is allowed, and Earth is clear.

► **A denser policy has a cost.**

- Testing all satellite pairs admits more candidates. Real systems still have finite terminals and pointing constraints.

Candidates become a time-specific graph



$$N = 72 : \quad N(N - 1)/2 = 2556 \text{ pairs}$$

All-pairs count excludes self-links and duplicates

SatNet Edu supports orbital-neighbor and distance policies. It does not allocate real antenna terminals.

Represent Each Snapshot as a Weighted Graph

Once usable links are known, a routing question can be expressed as a graph problem.

► **Vertices represent network objects.**

- Write $G(t) = (V, E(t))$. Satellites and ground endpoints are vertices. Each usable link is an edge at time t .

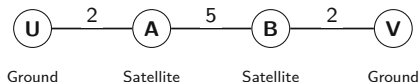
► **Weights express the objective.**

- Unit weights count hops. Propagation weights use distance divided by signal speed. Different weights can select different paths.

► **Respect forwarding permissions.**

- In SatNet Edu, a ground station may start or end a route. It cannot be an intermediate satellite-to-satellite relay.

From links to an adjacency table



U: A (2 ms)
A: U (2 ms), B (5 ms)
B: A (5 ms), V (2 ms)
V: B (2 ms)

One route: 3 links, $2 + 5 + 2 = 9$ ms

Constructed graph with undirected links. Edge labels are one-way propagation milliseconds.

The Fewest Hops Need Not Give the Least Delay

A route with fewer links may still travel a longer total distance.

► **Compare complete route sums.**

- The upper path A–B–D uses two links and costs 12 ms. The lower path uses three links and costs 9 ms.

► **Choose a metric before optimizing.**

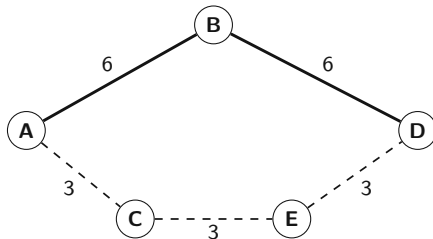
- Minimum hops selects the upper path. Minimum propagation delay selects the lower path in this graph.

► **Keep the claim specific.**

- A minimum-propagation route need not minimize real packet latency when queues, processing, or transmission times differ.

Two objectives, two answers

All edge weights are propagation delays in ms



Hops: A–B–D

Delay: A–C–E–D

Constructed weighted graph. All edges have nonnegative propagation delay. No queueing model.

Dijkstra Expands the Cheapest Unfinished Route

Nonnegative link delays let us settle the smallest tentative distance first.

► **Start at the source.**

- Set distance to A to zero and all others to infinity. Visiting A gives tentative costs B=6 and C=3.

► **Relax each outgoing edge.**

- From a settled vertex u , replace a larger estimate for v with $d(u) + w(u, v)$ and remember the predecessor.

► **Stop when the target is settled.**

- This graph reaches D at 9 ms through E. A discovered route is only tentative until it wins the comparison.

A worked shortest-path calculation

Edges (ms): AB=6, BD=6, AC=3, CE=3, ED=3

Settled	B	C	E	D
A (0)	6	3	∞	∞
C (3)	6	3	6	∞
B (6)	6	3	6	12
E (6)	6	3	6	9
D (9)	6	3	6	9

Predecessors: $D \leftarrow E \leftarrow C \leftarrow A$

$$d(v) \leftarrow \min\{d(v), d(u) + w(u, v)\}$$

Costs in ms. B is expanded before E for this equal-cost tie. Either order gives the same minimum cost.

Coverage at Both Ends Does Not Prove Connectivity

A route requires a continuous chain through the graph, not merely a first and last hop.

► **A missing bridge can split the graph.**

- Both terminals may keep access links while the only connection between their satellite groups disappears.

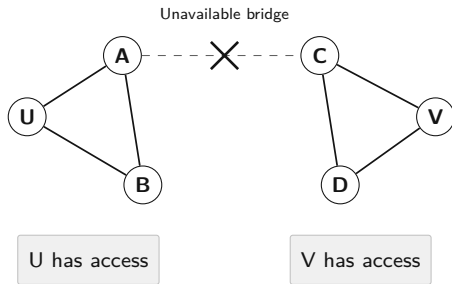
► **An alternate chain creates resilience.**

- A second route helps only if it avoids the failed links or nodes. Two paths can share the same vulnerable component.

► **Represent no path explicitly.**

- An unreachable sample has no valid delay. Replacing it with zero would make a disconnected design look fast.

Both sites covered, yet no end-to-end path



Constructed graph. Solid lines are usable. The dashed bridge is absent from the routing graph.

Compare Routing Objectives on the Same Snapshot

Use one recorded geometry to isolate what the route metric changes.

► Hold the snapshot fixed.

- Query the same source, target, and time. Changing time as well as metric would mix two causes.

► Inspect the selected paths.

- Compare node sequences, hop counts, and propagation sums. Equal answers are possible and are worth explaining.

► Check reachability first.

- If no path exists, inspect access links and connected groups before comparing delay values.

Two queries, one network state

```
snapshot = net.at(120)
for metric in ("hops", "delay"):
    route = snapshot.route(
        "A", "B", metric=metric)
    if route.reachable:
        print(metric, route.path)
        print(route.hops,
              route.propagation_ms)
    else:
        print(metric, route.reason)
```

Question: which extra links make a shorter route?

Run with the companion `routing_experiment.py`. SatNet Edu optimizes snapshots, not a distributed routing protocol.